1-DP-Playing with Numbers

Started on	Friday, 25 September 2026, 8:32 AM
State	Finished
Completed on	Friday, 25 September 2026, 8:35 AM
Time taken	2 mins 42 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 10.00 out of 10.00

Flag question

Question text

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer ‘n’ and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*
1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways ‘n’ can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	6	6	6	
	25	8641	8641	
	100	24382819596721629	24382819596721629	

Passed all tests!

Correct

Marks for this submission: 10.00/10.00.

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1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways ‘n’ can be represented using 1 and 3

Sample Input

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Sample Output

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Answer:(penalty regime: 0 %)

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2-DP-Playing with chessboard

Started on	Friday, 11 September 2026, 8:30 AM
State	Finished
Completed on	Friday, 25 September 2026, 8:40 AM
Time taken	14 days

Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 10.00 out of 10.00

Flag question

Question text

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:
Input
3
1 2 4
2 3 4
8 7 1
Output:
19

Explanation:
Totally there will be 6 paths among that the optimal is
Optimal path value:1+2+8+7+1=19

Input Format
First Line contains the integer n
The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	3 1 2 4 2 3 4 8 7 1	19	19	
	3 1 3 1 1 5 1 4 2 1	12	12	
	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	

Passed all tests!

Correct

Marks for this submission: 10.00/10.00.

3-DP-Longest Common Subsequence

Started on	Friday, 11 September 2026, 8:50 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:14 AM
Time taken	24 mins 44 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	aab azb	2	2	

	Input	Expected	Got	
	ABCD ABCD	4	4	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

4-DP-Longest non-decreasing Subsequence

Started on	Friday, 11 September 2026, 9:15 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:22 AM
Time taken	7 mins 32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

Question text

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer:(penalty regime: 0 %)

Feedback

	Input	Expected	Got	
	9 -1 3 4 5 2 2 2 2 3	6	6	
	7 1 2 2 4 5 7 6	6	6	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.